Chapter 1 Variational Autoencoder (VAE)

Introduction

- *Deep generative models*
- *Autoencoder (AE)*
- *Latent-variable generative modeling*
- *Variational inference and ELBO*
- *VAE objective*
- *Reparameterization trick*

In the previous chapter, we have already seen latent-variable models such as Mixture of Gaussians (MoG), whose generative story is explicit and simple. However, realistic data distributions (e.g. natural images) are far more complex:

- Images live in an extremely high-dimensional space (e.g. a 256×256 RGB image is a vector in $\mathbb{R}^{256 \cdot 256 \cdot 3}$);
- The distribution contains rich multi-scale structures (low-frequency layout, high-frequency texture, object semantics, etc.);
- A mixture of a small number of simple components (e.g. Gaussians) is usually not expressive enough.

This motivates *deep generative models*, which use neural networks to build highly expressive families of distributions.

Even if a distribution admits a closed-form density, sampling can still be hard. In practice, most sampling procedures ultimately rely on generating pseudo-random numbers (often starting from a nearly-uniform source on $[0, 1]$) and applying transformations. For complex high-dimensional distributions, sampling often requires Monte Carlo / MCMC methods, which may have large variance and long burn-in.

To alleviate the difficulty of directly sampling from complex high-dimensional distributions, we seek to learn a lower-dimensional representation in which the data distribution becomes easier to model and manipulate. This motivates the use of autoencoders.

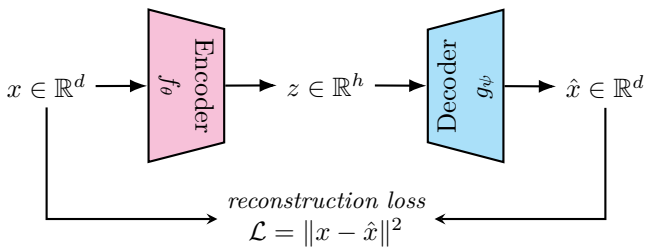
1.1 Autoencoders (AE)

An *autoencoder* maps a high-dimensional data point $x \in \mathbb{R}^d$ into a low-dimensional representation $z \in \mathbb{R}^h$ and reconstructs x from z, typically via an encoder–decoder pair:

$$z = f_\theta(x), \qquad \hat{x} = g_\psi(z). \tag{1.1}$$

Here $f_\theta : \mathbb{R}^d \to \mathbb{R}^h$ is called the **encoder** and $g_\psi : \mathbb{R}^h \to \mathbb{R}^d$ the **decoder**. The training objective minimizes a reconstruction error, e.g. mean squared error:

$$\operatorname*{argmin}_{\theta,\psi} \sum_{i=1}^{n} \|x_i - \hat{x}_i\|^2, \qquad \hat{x}_i = g_\psi(f_\theta(x_i)). \tag{1.2}$$



The learned representation z can be used for downstream tasks such as dimensionality reduction, clustering, or as a feature input to a classifier. **However, an AE is *not* a generative model**: it provides no principled distribution on z to sample from, so generating a new realistic x is not guaranteed.

1.2 Latent-Variable Generative Modeling

We now move from representation learning to generative modeling. Let $\mathcal{D} = \{x_i\}_{i=1}^n$ be an unlabeled dataset (e.g. images). Our goal is to learn a model that can generate *new* data similar to the training distribution. That is, we want to learn(or equivalently, approximate) a model $p_\theta(x)$ that can generate new data similar to the training distribution.

The key idea is to introduce a latent variable z with a simple prior $p(z)$, and generate x via a conditional distribution $p_\theta(x \mid z)$ whose parameters are produced by a neural network.

Definition 1.1 (Latent-variable generative model)

Choose a prior $p(z)$ (easy to sample), and a conditional likelihood $p_\theta(x \mid z)$ (expressive). The marginal model is

$$p_\theta(x) = \int p_\theta(x \mid z)\, p(z)\, \mathrm{d}z. \tag{1.3}$$

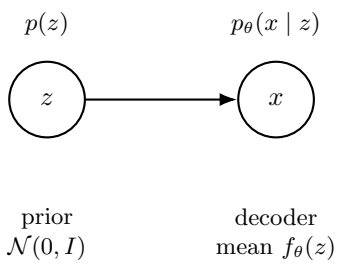
♣

Note

That is, $p_\theta(x)$ is the integral of $p_\theta(x \mid z)$ with measure $p(z)$.

A canonical choice is the standard Gaussian prior $p(z) = \mathcal{N}(0, I)$, and an isotropic Gaussian likelihood whose mean is a neural network:

$$p_\theta(x \mid z) = \mathcal{N}\big(x \mid f_\theta(z),\, \sigma^2 I\big). \tag{1.4}$$



Note

It is absolutely not necessary to use a Gaussian likelihood. Any distribution that is not ill-conditioned is suitable. Here we use a Gaussian likelihood only for simplicity.

Remark

If f_θ were linear, then x would remain Gaussian. A nonlinear and expressive f_θ enables highly non-Gaussian and complex distributions over x.

The maximum-likelihood objective is

$$\operatorname*{argmax}_\theta \sum_{i=1}^n \log p_\theta(x_i). \tag{1.5}$$

But expanding (1.3) gives

$$\log p_\theta(x) = \log \int p_\theta(x \mid z)\, p(z)\, \mathrm{d}z, \tag{1.6}$$

where the integral is typically intractable when $p_\theta(x \mid z)$ is defined by a deep neural network. One may try the Monte

Carlo approximation

$$p_\theta(x) \approx \frac{1}{K}\sum_{k=1}^{K} p_\theta\big(x \mid z^{(k)}\big), \qquad z^{(k)} \sim p(z), \tag{1.7}$$

i.e., estimate the marginal likelihood by sampling latent codes from the prior and averaging the corresponding conditional likelihoods. In principle this is valid, but in high dimensions most prior samples $z^{(k)}$ will fall in regions that are unlikely to explain a specific datapoint x. Consequently, the terms $p_\theta(x \mid z^{(k)})$ are typically extremely small for most samples, and the average is dominated by rare lucky valid samples. This leads to very large variance, so obtaining a reliable estimate would require a large K for each datapoint x_i, which quickly becomes computationally prohibitive.

Since this is a latent-variable model, one may also consider the EM algorithm. Recall that EM alternates between (i) inferring the latent variable distribution given the current parameters (E-step) and (ii) maximizing the expected complete-data log-likelihood under that inferred distribution (M-step). Concretely, the E-step requires the posterior

$$p_\theta(z \mid x) = \frac{p_\theta(x \mid z)p(z)}{p_\theta(x)} \propto p_\theta(x \mid z)p(z), \tag{1.8}$$

because the M-step depends on expectations taken with respect to $p_\theta(z \mid x)$. However, the normalization constant is exactly the marginal likelihood $p_\theta(x) = \int p_\theta(x \mid z)p(z)\,dz$, which is the same intractable integral as in (1.6). In classical mixture models (e.g., MoG), the latent variable is discrete, so the corresponding normalization is a finite sum and the posterior can be computed exactly; in VAE, z is continuous and high-dimensional, so the required integrals are generally intractable.

The essential bottleneck, therefore, is that we cannot evaluate or sample from the true posterior $p_\theta(z \mid x)$ efficiently. Variational inference addresses this by replacing the intractable posterior with a tractable approximation

$$q_\phi(z \mid x) \approx p_\theta(z \mid x), \tag{1.9}$$

where q_ϕ is chosen from a convenient family (e.g., Gaussians with diagonal covariance) and its parameters are produced by another neural network, called the *encoder*. Intuitively, instead of performing an exact E-step, we learn an inference model that takes a datapoint x as input and outputs a distribution over plausible latent codes z that could have generated x under the decoder.

Remark

Actually, sampling a random number itself is not easy. Even producing a single sample that is close to uniform on the interval $[0, 1]$ already relies on carefully designed pseudo-random number generators and deterministic transformations. Sampling from a more complex distribution, especially one defined on a high-dimensional space or on a complicated manifold, is therefore substantially more challenging.

Theorem 1.1 (ELBO decomposition)

For any distribution $q_\phi(z \mid x)$,

$$\log p_\theta(x) = \underbrace{\mathbb{E}_{z\sim q_\phi(\cdot\mid x)}\big[\log p_\theta(x \mid z)\big] - \mathrm{KL}\big(q_\phi(z \mid x) \,\big\|\, p(z)\big)}_{\mathcal{L}_{\mathrm{ELBO}}(x;\theta,\phi)} + \underbrace{\mathrm{KL}\big(q_\phi(z \mid x) \,\big\|\, p_\theta(z \mid x)\big)}_{\geq 0}. \tag{1.10}$$

Hence $\mathcal{L}_{\mathrm{ELBO}}(x;\theta,\phi) \leq \log p_\theta(x)$.

♡

Remark

In (1.10), the two KL divergences are different and should not be confused. The KL term inside the ELBO compares $q_\phi(z \mid x)$ with the prior $p(z)$, while the remaining KL term measures the gap between $q_\phi(z \mid x)$ and the true posterior $p_\theta(z \mid x)$.

Remark

The KL term $\mathrm{KL}(q_\phi(z \mid x) \| p(z))$ appears inside *ELBO and acts as a regularizer toward the prior. The KL term* $\mathrm{KL}(q_\phi(z \mid x) \| p_\theta(z \mid x))$ *is the* gap *between ELBO and the true log-evidence* $\log p_\theta(x)$.

During optimization, we optimize $\log p_\theta(x)$ indirectly by maximizing a tractable lower bound (the ELBO). (Because the gap is intractable)

Although the gap term $\mathrm{KL}\big(q_\phi(z \mid x) \,\big\|\, p_\theta(z \mid x)\big)$ cannot be optimized explicitly, both q_ϕ and p_θ depend on the parameters (ϕ, θ) and are updated jointly when maximizing the ELBO. As a result, the gap is implicitly affected during training and does not remain fixed or uncontrolled.

The expectation in $\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)]$ still involves an integral, but the sampling distribution is now data-dependent. This is a major improvement: the most costly part in practice is drawing informative samples, and sampling from the prior is inefficient. By replacing prior sampling with sampling conditioned on the observed data, the computation is focused on latent regions that actually explain x. Instead of drawing samples from the broad prior $p(z) = \mathcal{N}(0, I)$, we sample from the posterior $q_\phi(z \mid x)$, which typically concentrates its mass in a narrow region relevant to the given data point. As a result, Monte Carlo estimates of the expectation can have significantly lower variance.

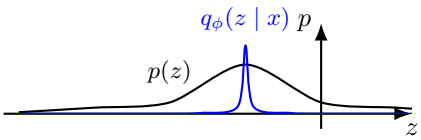


Figure 1.1: A narrow $q_\phi(z \mid x)$ concentrates its mass in a relatively narrow region, while a broad $p(z)$ has a wide distribution.

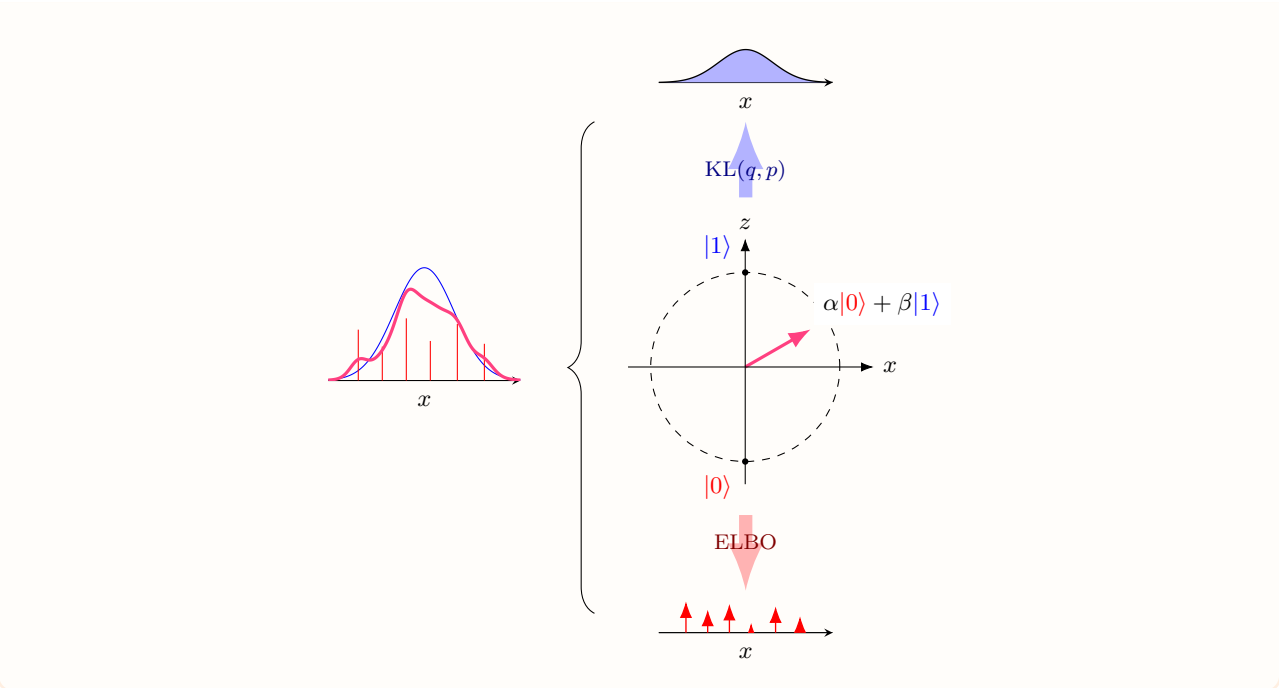
Note

From the perspective of information theory, the condition x brings additional information about z, which increase the success probability of sampling a valid, meaningful z.

Note

The prior $p(z)$ *does not participate in training by being sampled in the forward optimization path. Instead, it enters training through the KL regularization term, which constrains and aligns the overall shape of* $q_\phi(z \mid x)$. *Meanwhile, by sampling* $z \sim q_\phi(z \mid x)$ *and reconstructing x, the model is forced during self-optimization to maintain stable decoding within a neighborhood around* $\mu_\phi(x)$, *thereby endowing the latent representation with semantic consistency and smoothness.*

The weight of the KL term must be balanced. If the KL term is too small, $q_\phi(z \mid x)$ *tends toward deterministic encoding* ($\sigma \to 0$), *and the latent space lacks a well-structured global organization, leading to poor interpolation and unreliable sampling from the prior. If the KL term is too large,* $q_\phi(z \mid x)$ *is pushed close to* $p(z)$, *and the latent variables carry insufficient information about x, making it difficult for the decoder to recover meaningful samples. These two cases correspond respectively to an irregular latent space and information collapse.*



Training VAE maximizes the ELBO over the dataset:

$$\underset{\theta,\phi}{\operatorname{argmax}} \sum_{i=1}^{n} \mathcal{L}_{\mathrm{ELBO}}(x_i;\theta,\phi). \tag{1.11}$$

Assume the decoder likelihood (1.4). Then, up to an additive constant independent of θ,

$$\log p_\theta(x \mid z) = -\frac{1}{2\sigma^2}\left\|x - f_\theta(z)\right\|^2 + C. \tag{1.12}$$

Plugging this into (1.10), maximizing ELBO is equivalent to minimizing (ignoring constants)

$$\underset{\theta,\phi}{\operatorname{argmin}} \sum_{i=1}^{n} \left[\mathbb{E}_{z\sim q_\phi(\cdot\mid x_i)}\left\|x_i - f_\theta(z)\right\|^2 + \beta\,\mathrm{KL}\big(q_\phi(z \mid x_i)\,\big\|\,p(z)\big)\right], \tag{1.13}$$

where $\beta = 2\sigma^2$ (or we may treat β as a tunable weight, leading to β-VAE).

Remark

If we remove the KL regularization term, (1.13) becomes a standard autoencoder objective: it reconstructs well but does not enforce a tractable sampling distribution in latent space, hence is not a principled generative model.

The KL regularizer encourages $q_\phi(z \mid x)$ to be close to the fixed prior $p(z) = \mathcal{N}(0, I)$. After training, to *generate* a new sample, we do not need the encoder:

$$z \sim p(z), \qquad x \sim p_\theta(x \mid z). \tag{1.14}$$

Because $p(z)$ is simple to sample and the decoder is learned, we can generate new x.

Note

In modern diffusion systems, VAE is often used as a latent autoencoder*, diffusion operates in the latent space, then the VAE decoder maps latent variables back to pixel space.*

1.3 Reparameterization Trick

To optimize (1.13) with backpropagation, we must be able to compute gradients with respect to the encoder parameters ϕ. However, if we sample directly by $z \sim q_\phi(z \mid x)$, then z is produced by a stochastic sampling operation rather than an explicit differentiable function of ϕ. In other words, small changes in ϕ change the distribution $q_\phi(z \mid x)$, but the random draw z itself does not provide a differentiable path for gradients to flow back to ϕ, which breaks end-to-end training.

A standard choice is to let the encoder output a Gaussian distribution

$$q_\phi(z \mid x) = \mathcal{N}\big(z \mid \mu_\phi(x),\, \Sigma_\phi(x)\big), \tag{1.15}$$

often with diagonal covariance

$$\Sigma_\phi(x) = \operatorname{diag}\big(\sigma_\phi^2(x)\big). \tag{1.16}$$

Sample instead from a parameter-free noise distribution:

$$\varepsilon \sim \mathcal{N}(0, I), \qquad z = \mu_\phi(x) + L_\phi(x)\,\varepsilon, \tag{1.17}$$

where $L_\phi(x) L_\phi(x)^\top = \Sigma_\phi(x)$ (for diagonal covariance, $L_\phi(x) = \operatorname{diag}(\sigma_\phi(x))$).

Proposition 1.1 (Correctness of reparameterization)

If $\varepsilon \sim \mathcal{N}(0, I)$ and $z = \mu + L\varepsilon$, then $z \sim \mathcal{N}(\mu, LL^\top)$.

Proof

The affine transformation of a Gaussian remains Gaussian. Moreover,

$$\mathbb{E}[z] = \mu + L\,\mathbb{E}[\varepsilon] = \mu, \tag{1.18}$$

and

$$\operatorname{Cov}(z) = L \operatorname{Cov}(\varepsilon)\, L^\top = LIL^\top = LL^\top. \tag{1.19}$$

The sampling step now occurs in ε, which has no learnable parameters; gradients only need to flow through differentiable operations producing $\mu_\phi(x)$ and $L_\phi(x)$.

Note

In a deterministic autoencoder, each data point x is mapped to a single latent code z, and the reconstruction loss only enforces correctness at that particular point. The decoder is therefore unconstrained in the neighborhood around z.

In contrast, a VAE represents each data point by a Gaussian distribution $q_\phi(z \mid x) = \mathcal{N}(\mu(x), \sigma^2(x))$. During training, latent variables are sampled from this distribution. As a result, the decoder is required to reconstruct x not just at a single latent code, but throughout a neighborhood around $\mu(x)$. This stochasticity induces a smooth latent space.

The KL term in (1.13) is tractable when both distributions are Gaussian.

Theorem 1.2 (Gaussian–Gaussian KL)

Let $q = \mathcal{N}(\mu, \Sigma)$ and $p = \mathcal{N}(0, I)$. Then

$$\mathrm{KL}(q \| p) = \frac{1}{2}\Big(\operatorname{tr}(\Sigma) + \mu^\top \mu - d - \log\det(\Sigma)\Big). \tag{1.20}$$

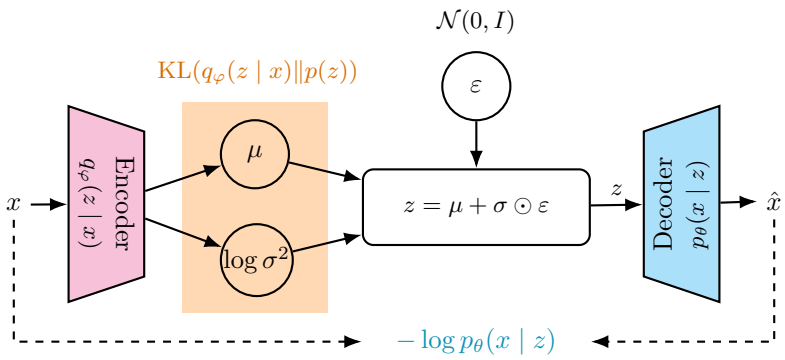


Figure 1.2: The structure of a VAE, in which orange KL loss is regularized towards the prior $p(z)$ which is trained for sampling during generation, and the cyan ELBO loss is regularized towards the decoder $p_\theta(x \mid z)$. The source of randomness is shifted from sampling z itself to sampling an auxiliary variable ε, so that z becomes a deterministic function of ε and the parameters. As a result, the gradient paths through the functions of z in both p_θ and q_ϕ are preserved.

In the diagonal case $\Sigma = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_d^2)$*, this becomes*

$$\mathrm{KL}(q \| p) = \frac{1}{2} \sum_{j=1}^{d} \left(\mu_j^2 + \sigma_j^2 - 1 - \log \sigma_j^2 \right). \tag{1.21}$$

Remark

Therefore, VAE training typically uses:

- *a decoder network* f_θ *for* $p_\theta(x \mid z)$*;*
- *an encoder network producing two output heads:* $\mu_\phi(x)$ *and* $\log \sigma_\phi^2(x)$*;*
- *one (or a few) Monte Carlo samples per datapoint for the reconstruction expectation.*